

## 1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

**2. EXECUTION ENVIRONMENT OVERVIEW** ComicCraft operates as an asynchronous Python web application using the FastAPI framework. Since it offloads heavy Generative AI processing (Google Gemini and Hugging Face) to cloud endpoints, it does not require a complex local database or a compiled .exe installer. The software is executed directly from a lightweight Python Virtual Environment, ensuring seamless cross-platform compatibility.

## 3. SYSTEM REQUIREMENTS

- **Language Setup:** Python 3.10 (or a newer stable release) must be installed.
- **Operating Environment:** Windows OS (10/11) is recommended, though it functions perfectly on Unix-based systems with equivalent bash commands.
- **Network Connectivity:** A stable, unrestricted internet connection is mandatory to fetch AI responses in real-time.

## 4. APPLICATION SETUP & RUNTIME INSTRUCTIONS

**Step 1: Access the Project Directory** Extract the project files and open your command line interface (Terminal or PowerShell) directly inside the root folder (where `main.py` and `requirements.txt` are located).

**Step 2: Initialize Virtual Environment** To prevent dependency conflicts with other local software, create and activate an isolated environment:

```
python -m venv env
.\env\Scripts\activate
```

**Step 3: Install Required Dependencies** Populate the environment with all necessary third-party libraries (e.g., `fastapi`, `uvicorn`, `fpdf`, `pillow`) by executing:

```
pip install -r requirements.txt
```

**Step 4: API Key Configuration** Create a hidden file named `.env` in the root directory. Insert your active cloud AI credentials as follows to authorize the application:

```
GEMINI_API_KEY=insert_your_google_api_key_here  
HF_API_KEY=insert_your_huggingface_api_key_here
```

**Step 5: Boot the Application Server** Launch the local Uvicorn ASGI server to host the frontend interface:

```
python -m uvicorn app.main:app --reload
```

**Step 6: Launch the Interface** Open your preferred web browser and navigate to the local host address to begin using ComicCraft:

<http://127.0.0.1:8000>

**5. ERROR HANDLING & DEVELOPMENT FIXES** During local execution, specific system restrictions might block the setup. The following workarounds were implemented during development:

- **Execution Policy Block (Activate .ps1 Error):** Windows PowerShell often blocks local scripts by default. This is resolved by running `Set-ExecutionPolicy RemoteSigned -Scope CurrentUser` before attempting to activate the env.
- **Path Length Exception (WinError 206):** Installing massive Python packages can exceed Windows directory limits. Fixed by updating the Registry (`LongPathsEnabled = 1`).

- **Missing Package Error (ModuleNotFoundError):** If the server fails to boot, it indicates the `requirements.txt` did not install correctly. Restart the terminal, ensure `(env)` is visible, and reinstall the packages.
- **Network Connection Refused (getaddrinfo failed):** If API calls fail due to DNS errors, the system is designed to automatically bypass the error and generate a yellow PIL-based placeholder image locally, preventing a total crash.